



# NORTH SOUTH UNIVERSITY

School of Engineering and Physical Sciences

Department of Electrical and Computer Engineering

CSE482L.1: Internet and Web Technologies Lab

## Project Proposal

### ResumeSync: AI-Powered ATS-Friendly Resume Builder & Analyzer

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# 1 Project Title and Objective

## 1.1 Project Title

ResumeSync: AI-Powered ATS-Friendly Resume Builder & Analyzer

## 1.2 Project Objective

The primary objective of this project is to develop a comprehensive, one-stop web platform that empowers job seekers to create, analyze, and optimize high-quality resumes, specifically engineered to meet and exceed the stringent requirements of modern Applicant Tracking Systems (ATS).

At its core, the project will integrate a sophisticated evaluation system powered by multiple high-end Large Language Models (LLMs) accessed via APIs. This AI-driven engine will go beyond basic template formatting to:

- **Provide instant, data-driven feedback** on ATS compatibility, content clarity, and keyword optimization.
- **Offer intelligent suggestions** for improving resume organization and structure.
- **Deliver actionable recommendations** for continuous improvement, helping users refine their content to be more impactful.

Ultimately, ResumeSync aims to be more than just a builder; it will serve as an intelligent career tool that provides users with the expert guidance needed to navigate the automated recruitment landscape successfully, bridging the gap between their qualifications and their ability to get noticed.

## 2 Problem Statement

In today's job market, most large companies use Applicant Tracking Systems to filter job applicants. It's estimated that over 98% of Fortune 500 companies rely on this technology. These systems automatically reject up to 75% of resumes before a human ever sees them [1]. This creates major problems for job seekers:

- **Parsing Errors:** Resumes with creative designs, like tables or graphics, can confuse an ATS, leading to rejection based on formatting, not skills.
- **Poor Keyword Matching:** ATS software ranks resumes based on how well they match keywords from the job description. Many applicants don't know how to include the right keywords, so they get ranked poorly.
- **Lack of Expert Feedback:** Without immediate advice, applicants keep submitting flawed resumes, leading to a long and frustrating job search.

Our project aims to provide a simple solution to these technical problems, helping good candidates get noticed.

### 3 Proposed Solution Overview

ResumeSync will be a website that combines an easy-to-use resume builder with a powerful AI analysis tool. The application will help users create a resume that is both ATS-friendly and impressive to recruiters.

#### Key Features

- **User Accounts:** Secure registration and login for users to save and manage their work.
- **Interactive Resume Builder:** A guided form to easily input all the standard parts of a resume (contact info, experience, etc.).
- **ATS-Friendly Templates:** A library of clean, professional templates designed to be read perfectly by an ATS.
- **Live Preview:** A real-time preview that shows how the resume looks as the user types.
- **AI-Powered Analysis:** Using the Gemini API, our key feature will analyze a user's resume against a job description to provide:
  - An **ATS Compatibility Score** to quickly see how well it matches.
  - **Keyword Suggestions** by identifying important words from the job post that are missing from the resume.
  - **Content Improvement Tips**, like using stronger verbs or better describing achievements.
- **PDF Export:** The ability to download the final resume as a clean PDF, ensuring it looks the same everywhere.

### 4 Target Users / System Users

- **Primary Users:**
  - **Job Seekers:** Anyone applying for jobs who wants a better chance of success.
  - **Recent Graduates:** Students and new graduates who need help creating their first professional resume.
  - **Career Changers:** People switching industries who need to highlight their transferable skills.
- **User Needs:**
  - A simple tool that doesn't require any design skills.
  - Assurance that their resume is correctly formatted for automated systems.
  - Helpful advice on how to make their resume content stronger.

## 5 Use Case Diagram

The main interactions between a user (Job Seeker) and the system are shown in the use case diagram below. This includes managing their account, creating and editing resumes, and using the AI analysis feature.

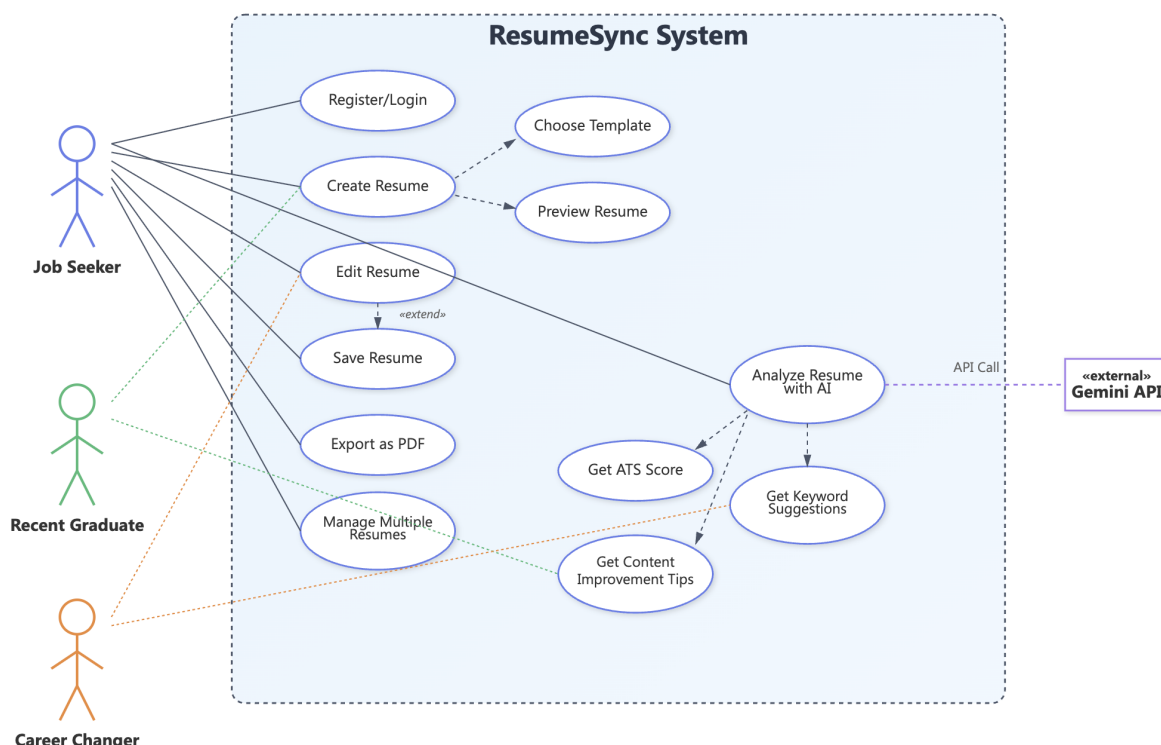


Figure 1: Use Case Diagram showing user interactions with the ResumeSync system.

## 6 User Stories

These user stories describe the needs of our target users and illustrate how they will interact with the ResumeSync platform to achieve their goals.

### As a General Job Seeker, I want to:

- Create a professional resume using **guided, step-by-step forms**, so that I can be confident the final document is structured correctly and meets professional standards.
- Instantly see an **ATS compatibility score and a keyword analysis** against a specific job description, so that I can understand my resume's effectiveness for a role *before* I apply.
- Receive specific, AI-powered suggestions on **how to rephrase my bullet points with stronger action verbs**, so that my achievements sound more impactful to recruiters.

- Save and manage **multiple versions of my resume**, so that I can easily tailor and optimize each one for different job applications without starting from scratch.
- Download my final resume in a **clean, universally compatible PDF format**, so that I can submit it anywhere without worrying about formatting errors.

### As a Career Changer, I want to:

- Get AI-powered recommendations on **how to reframe my past experience**, so that it highlights the transferable skills that are most relevant to a new industry.
- Have the system **identify the key terminology and skills** from job descriptions in my target field, so that I can learn the industry's language and make my resume look like a perfect fit.

### As a Recent Graduate, I want to:

- Receive AI guidance on how to **structure my resume to emphasize my education, projects, and internships**, so that I can create a strong application even with limited professional experience.
- Use AI suggestions to **describe my academic projects and coursework**, so that they translate into real-world skills and accomplishments that impress employers.

## 7 Interface Sketch / Wireframe

Below are rough layouts for the main pages of the application to visualize the user experience.

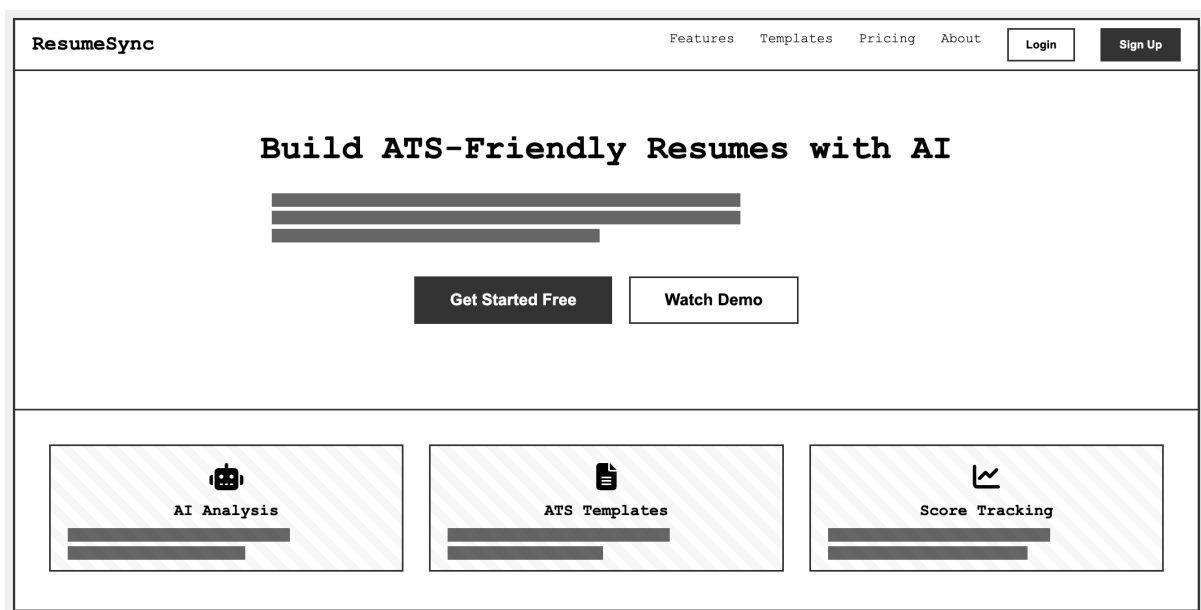


Figure 2: Wireframe sketch of the Home Page.

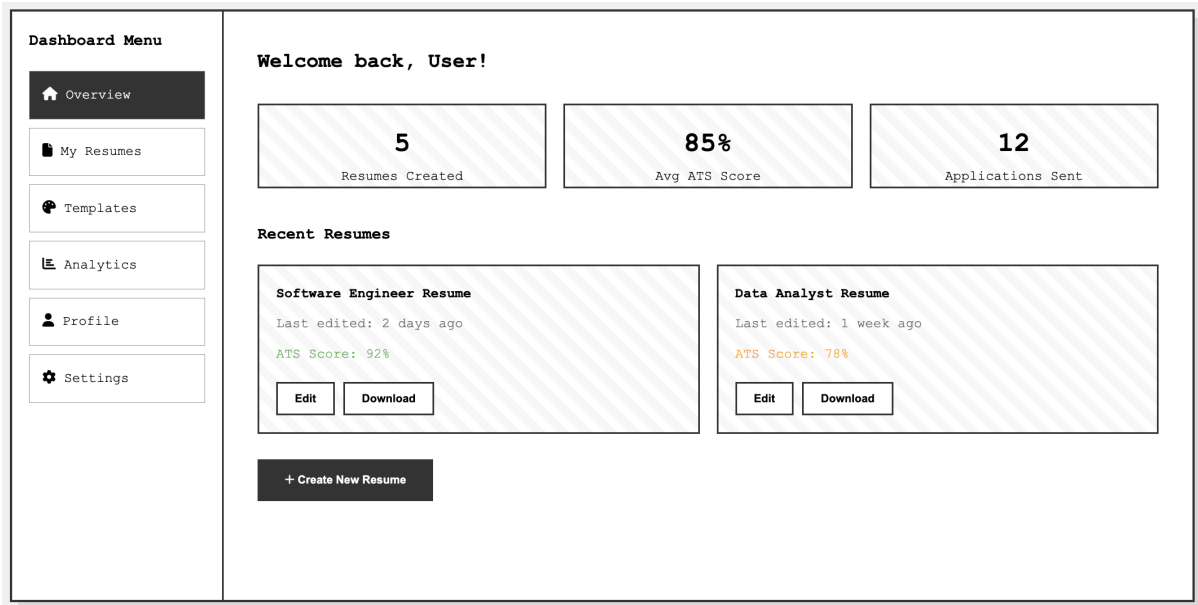


Figure 3: Wireframe sketch of the User Dashboard.

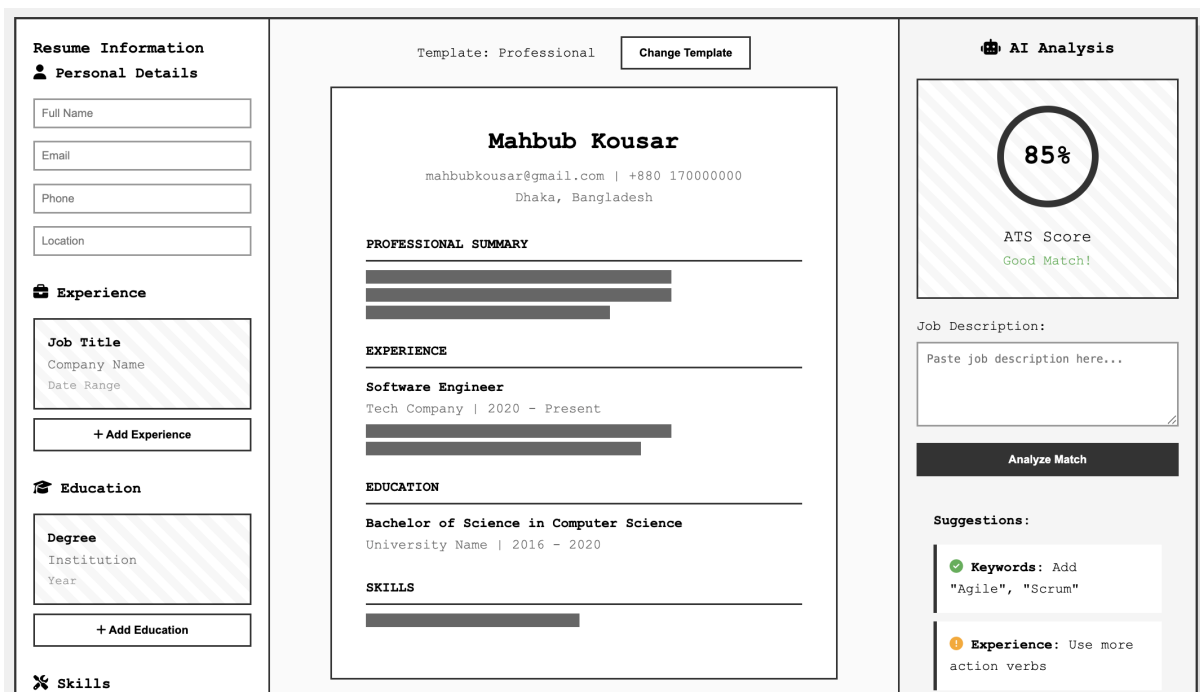


Figure 4: Wireframe sketch of the Resume Editor, showing the form, live preview, and AI analysis panel.

## 8 Technology Stack

Our project will be built using modern, reliable technologies that fit the course requirements.

- **Front-End:**
  - **HTML5:** For the structure of the web pages.

- **CSS3 & Bootstrap:** For styling and making the site responsive on all devices.
- **JavaScript & jQuery:** For user interactivity, form validation, and using AJAX to get AI analysis without reloading the page.
- **Back-End:**
  - **PHP 7.4+:** For server-side logic, managing users, and connecting to the database and external APIs.
  - **MySQL 5.7+:** For securely storing all user and resume data.
- **APIs and Integration:**
  - **Google Gemini API:** This will be the core of our smart features, used for analyzing resumes and providing feedback [2].
  - **FPDF (PHP Library):** A server-side library to generate the final PDF downloads.
- **Development and Hosting:**
  - **Environment:** We will use XAMPP for local development, VS Code as our editor, and Git for version control.
  - **Hosting:** The final project will be deployed on a live server like 000webhost or InfinityFree.

## 9 Additional Features & Potential Enhancements

While the core focus of our project is to deliver the robust, web-based AI resume platform as described, we have identified several advanced features that could be pursued if time allows. These are considered stretch goals that would extend the platform's capabilities to cater to a wider range of user preferences and workflows.

### 1. Conversational Resume Generation (Chat Interface)

To move beyond the limitations of traditional web forms, we envision a conversational AI interface. Instead of filling out static input boxes, users could interact with the platform in natural language. The AI would intelligently parse, categorize, and format this information, asking follow-up questions as needed to create a more intuitive and dynamic resume-building experience.

### 2. Headless Resume Management via the Model Context Protocol (MCP)

For power users and developers, we propose a headless architecture that leverages the **Model Context Protocol (MCP)**, a sophisticated protocol developed by Anthropic. This would allow for structured, multi-turn interactions with our AI engine directly from a Command-Line Interface (CLI), enabling workflows far more complex than a standard API could handle.



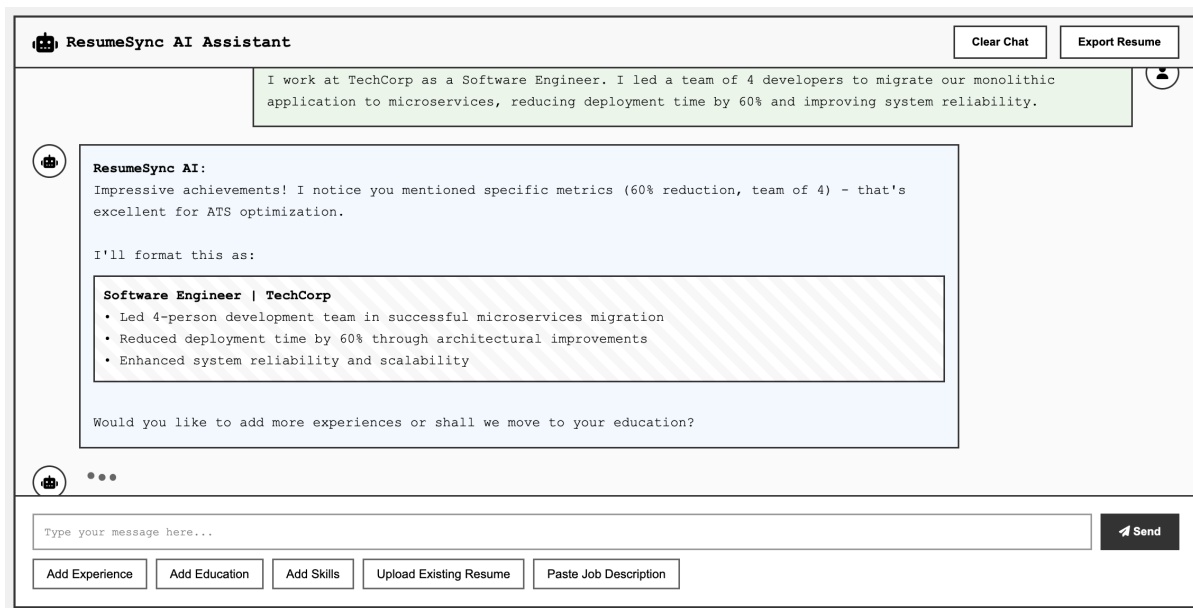


Figure 5: Conceptual mockup of the chat-driven resume generation feature.

**How it would work:** A backend server would host our core logic and implement the MCP. Users would interact with it via a lightweight local CLI client that “speaks” this protocol. This architecture is ideal for complex tasks like stateful analysis. For example, a user could run `resumesync create --from my_cv.pdf`. The server-side AI, communicating via MCP, could identify a vague bullet point and instruct the CLI to prompt the user for specific metrics, all while maintaining the context of the original request. This feature would provide a uniquely powerful and scriptable way to interact with our platform.

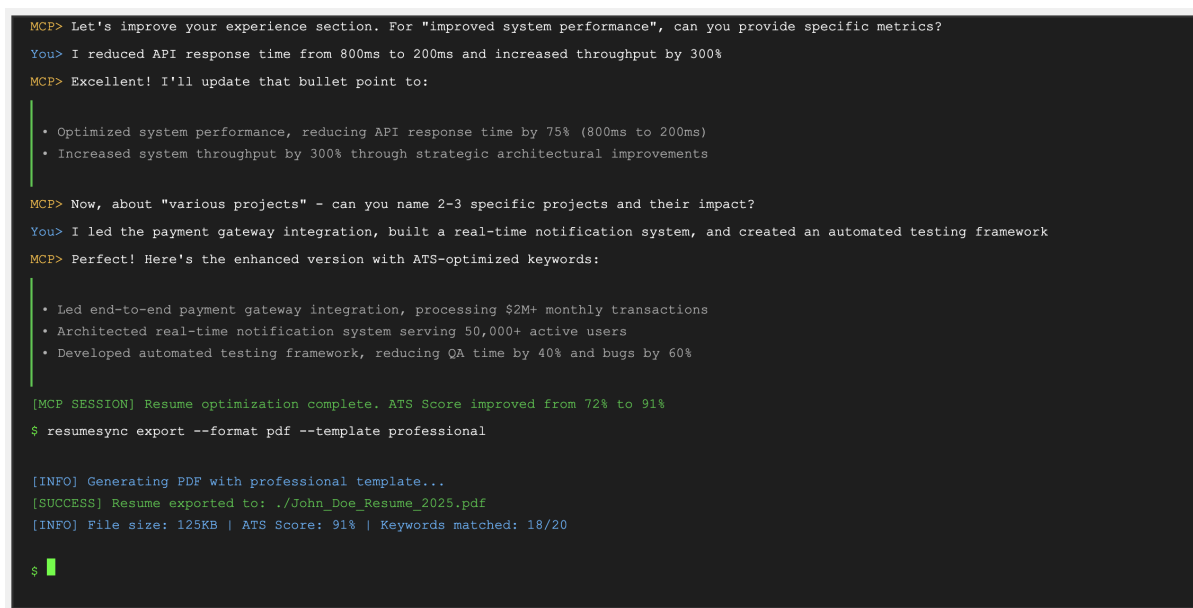


Figure 6: Conceptual mockup of the CLI tool interacting with the MCP server.

## References

- [1] Ikjyot Singh, Ankit Garg, and Deepti Sharma. “A Comprehensive Study of Resume Ranking Techniques and Their Applications”. In: *2025 12th International Conference on Computing for Sustainable Global Development (INDIACom)*. IEEE. 2025, pp. 1–6.
- [2] Timothy R McIntosh et al. “From google gemini to openai q\*(q-star): A survey on reshaping the generative artificial intelligence (ai) research landscape”. In: *Technologies* 13.2 (2025), p. 51.